# IMAD EDDINE EL MOUSS

Research Engineer | Applied AI, Knowledge Extraction & On-Premises Machine Learning

Turku, Finland • +358 41 574 6384 • imadeddine200507@gmail.com
imadlab.com • github.com/imaddde867 • linkedin.com/in/imad-eddine-e

## RESEARCH PROFILE

Research engineer at a Finnish applied-AI research group, working on machine learning systems that must run inside industrial environments under privacy, auditability and hardware constraints. My work sits between two questions: how unstructured industrial evidence (documents, speech, images, sensor streams) can be turned into structured, checkable representations, and how such systems should be evaluated so that a result means something outside the lab. Named inventor on a European patent application, contributor to a funded Horizon Europe consortium, and lead author of an applied paper in preparation. I am applying for a master's programme to deepen the methodological side of this work.

## EDUCATION

### B.Eng. ICT — Data Engineering & Artificial Intelligence

Jan 2023 – Dec 2025

*Turku University of Applied Sciences* — Turku, Finland

- Graduated 31 December 2025 with **243 ECTS** against a 240 ECTS programme. **Degree average 4.32/5**. **Thesis graded 5/5**.
- **Thesis:** *Local procedural knowledge graph extraction from safety-critical industrial manuals*. A privacy-by-design pipeline that runs entirely on local hardware and converts unstructured procedural documentation into knowledge graphs of steps, constraints and decision points. Contributions: hierarchy-aware semantic chunking, a two-stage extraction prompt structure, a purpose-built constraint-attachment metric, and a multi-seed evaluation harness with bootstrap confidence intervals and inter-annotator agreement gating.

## RESEARCH INTERESTS

Knowledge extraction and structured representation from unstructured and multimodal industrial data • on-premises and privacy-preserving machine learning under real hardware limits • evaluation methodology for language-model systems, including annotation quality, agreement and reproducibility • supervisory and adaptive control of deployed AI components • multimodal sensing for process monitoring.

## RESEARCH EXPERIENCE

### Research Engineer

Jan 2026 – Present

*Cognitive Technologies Research Group (CoRe), Turku University of Applied Sciences* — Turku, Finland

- Architect and principal developer of **AIOP**, an on-premises event-driven platform (Python, NATS JetStream, Docker, FastAPI) that carries every applied-AI pilot the group runs: immutable event envelope, plugin interface, single-writer state projector, per-site data isolation.
- Produced the platform's empirical performance record through a reproducible pair of stress audits — 200 events/s sustained at 0% loss with 10 ms p99 latency, measured ceiling ≈260 events/s — and removed an $O(N)$ state-staging bottleneck that would have capped throughput at ≈40 events/s at million-entity scale.
- Design and evaluate local language- and vision-model pipelines against real industrial data: quantised inference on RTX 5090/3090 workstations, schema-enforced structured extraction, and speech-to-report pipelines in which a hallucinated field is treated as an extraction failure.
- Research contributions across two funded programmes, **TeoÄly** (ERDF) and **ADINO** (Business Finland): machine-vision surface inspection, multi-channel spectral sensing for process control, document and ERP extraction, and speech-to-structured-report generation.
- Authored the group's **EU AI Act** position for its industrial offerings, including per-offering Annex III risk classification and the timeline as amended by Regulation (EU) 2026/1744.
- Supervise student research projects across a rolling cohort — more than twenty to date, several concurrent at any time, including thesis-track work — under a handoff standard requiring runnable setup, declared output schema, evaluation evidence and documented limitations.

### Applied AI Intern — Horizon Europe proposal work

Apr 2025 – Aug 2025

*Cognitive Technologies Research Group (CoRe), Turku University of Applied Sciences* — Turku, Finland

- Built a multi-source extraction prototype (OCR, document parsers, language models) producing schema-validated structured output from heterogeneous industrial documents.
- Contributed privacy-preserving data architecture, ontology alignment and multimodal harmonisation to a Horizon Europe research proposal.

### Research Intern — Autonomous & Intelligent Systems

Feb 2025 – Jun 2025

*Turku University of Applied Sciences* — Turku, Finland

- Time-series methods on production sensor data in InfluxDB: anomaly detection, forecasting and classification, alongside the ETL and reliability work the analysis depended on.

## RESEARCH OUTPUT & INTELLECTUAL PROPERTY

- **Patent (pending).** Named inventor, one of three, on a European patent application filed at the European Patent Office on 10 April 2026: adaptive digital content generation based on multimodal data. Applicant: Turku University of Applied Sciences. Publication pending.
- **Horizon Europe.** Contributor to the planning of *Sensor4Food* (HORIZON-CL4), funded at EUR 6.98M across a 21-partner European consortium coordinated by Turku UAS, developing photonic sensing and AI for the poultry value chain with pilots in Spain, Italy and Norway (2026–2030).
- **Paper in preparation (lead author).** Applied study on operating and supervising AI systems on-premises in manufacturing environments, evaluated against operational outcomes on a live deployment rather than on answer accuracy alone. Target: applied/in-use research venue, 2027.
- **Thesis artifact.** The procedural-knowledge-extraction pipeline, corpus tooling and evaluation harness are maintained as a reusable research codebase.

## SUPERVISION & TEACHING

- Technical supervisor within CoRe's student research programme — more than twenty projects to date on a rolling intake, covering 3D reconstruction, annotation tooling, speech-to-text, data anonymisation and ERP extraction — from project scoping and method choice through GPU environment provisioning, code review and final demonstrations.
- Built the group's onboarding and handoff practice for student researchers, including compute access on RTX 5090 workstations and national CSC computing resources.

## TECHNICAL SKILLS

| | |
|---|---|
| **Languages** | Python, SQL, JavaScript, Bash |
| **ML & Vision** | PyTorch, scikit-learn, TensorFlow, YOLO-family detectors, VLM prompting and LoRA/QLoRA fine-tuning, anomaly detection, forecasting |
| **LLM Systems** | llama.cpp, vLLM, Ollama, quantisation, schema-enforced extraction, RAG, knowledge graphs, Whisper STT, evaluation-harness design |
| **Data & Pipelines** | NATS JetStream, MQTT, Kafka, Airflow, ETL, PostgreSQL, InfluxDB, Neo4j, validation and row-level provenance |
| **Method** | Reproducible benchmarking, multi-seed evaluation, bootstrap confidence intervals, inter-annotator agreement, preregistration, EU AI Act risk classification |
| **Infrastructure** | Docker, FastAPI, Linux, Git, self-hosted CI, NVIDIA RTX 5090/3090 workstations, CSC national computing, Raspberry Pi and ESP32 edge hardware |

## CERTIFICATIONS, LANGUAGES & REFERENCES

| | |
|---|---|
| **Certifications** | AWS Academy — Cloud Foundations, Machine Learning Foundations, Machine Learning for NLP • DataCamp Certified Data Engineer • MATLAB Onramp & Fundamentals |
| **Languages** | Arabic (native) • Amazigh (native) • English (full professional) • French (native) • Finnish (A2) |
| **References** | Available on request, including the research group leader and the project lead at CoRe, Turku University of Applied Sciences |